

Tribological behavior of AA7075 hybrid composites reinforced with Al₂O₃ and TiO₂ particles

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The study investigates the wear behavior of AA7075 hybrid composites reinforced with varying percentages of Al₂O₃ and TiO₂. The microstructure reveals a homogenous distribution of reinforcing particles in the aluminum alloy matrix, enhanced by the addition of K2TiF6 flux. The wear process parameters, including % reinforcement, applied load, temperature, sliding velocity, and sliding distance, are systematically studied to understand their effects on wear resistance. The results reveal a non-linear trend in wear rate with increasing % reinforcement, showing an optimal percentage for the lowest wear rate. Applied load initially improves wear resistance but can lead to increased wear rates beyond threshold limit. The wear rate decreases initially with increasing temperature, indicating improved wear resistance, but increased at higher temperatures due to thermal softening effects. Sliding velocity and distance also significantly influence wear behavior, with lower wear rates observed at moderate velocities and distances due to the formation of protective layers. The addition of reinforcement reduces the COF, the relationship between COF and wear rate is inversely proportional, with lower COF leading to lower wear rates. The presence of reinforcement stabilizes the COF at different sliding distances, indicating improved wear resistance. The worn surface morphology indicates severe abrasion, micro-cutting, and plastic deformation, influenced by particle characteristics and load.

Keywords: Tribology, Friction, Surface morphology, Hybrid composites, Wear mechanisms.

Introduction

Aluminium matrix composites (AMCs) gaining importance due to their lightweight and excellent mechanical characteristics, making them suitable for a variety of applications. Hybrid composites, which are made by mixing various reinforcing materials, have the potential to improve the characteristics of AMCs [1, 2]. The wear behavior of these hybrid composites influenced by various factors, including the type and content of reinforcement materials, the matrix material, and the manufacturing process [3]. The wear process parameters, such as applied load, sliding speed, and sliding distance, plays a critical role in determining the wear behavior of hybrid composites [4, 5]. Increased applied load typically results in an increase in wear rate. Al7075-based hybrid composites reinforced with SiC and Al₂O₃ showed

that increasing the weight proportion of reinforcement increased wear resistance, with greater applied loads resulting in lower wear rates [6]. This behavior can be due to greater contact pressure and energy dissipation at higher loads, which causes more severe wear [7]. Augmented sliding speed increase wear rate owing to increased energy input and temperature increases at higher speeds. In magnesium-SiC-Gr hybrid composites, increasing the sliding speed resulted in increased wear rates, which were ascribed to the higher temperatures generated at higher speeds, which can accelerate wear processes such as abrasion and adhesion [8].

Increased sliding distance can cause an increase in wear rate, owing to the accumulation of wear debris and changes in contact conditions over time. AMCs reinforced with SiC and Al₂O₃ reported that the 15% hybrid composite exhibited superior wear resistance compared to the 5% composite which revealed that the type and concentration of reinforcement material can significantly influence wear properties [9]. Al7150-hBN nanocomposites found that the wear resistance of the

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